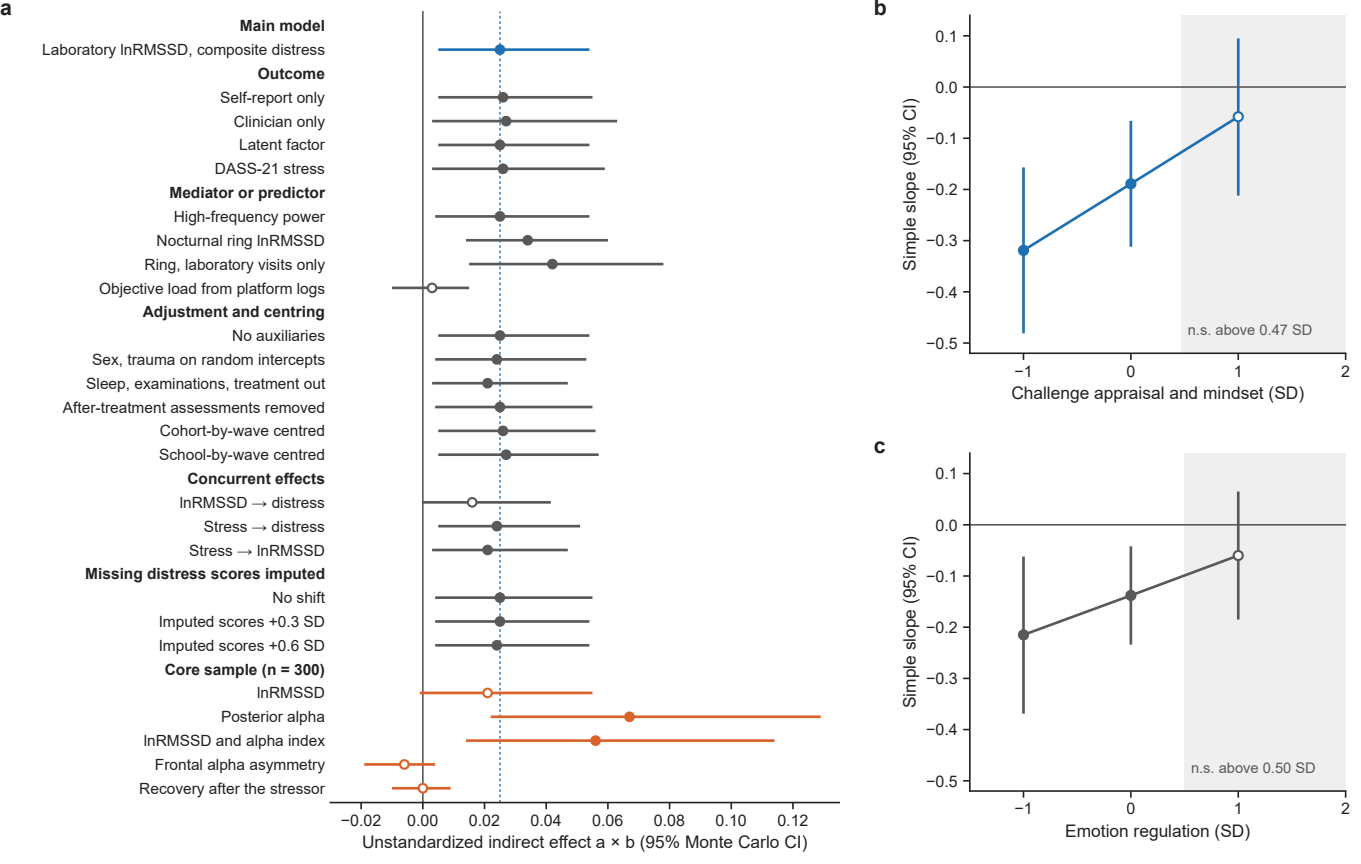

a
Main model
Laboratory lnRMSSD, composite distress
Outcome
Self-report only
Clinician only
Latent factor
DASS-21 stress
Mediator or predictor
High-frequency power
Nocturnal ring lnRMSSD
Ring, laboratory visits only
Objective load from platform logs
Adjustment and centring
No auxiliaries
Sex, trauma on random intercepts
Sleep, examinations, treatment out
After-treatment assessments removed
Cohort-by-wave centred
School-by-wave centred
Concurrent effects
lnRMSSD → distress
Stress → distress
Stress → lnRMSSD
Missing distress scores imputed
No shift
Imputed scores +0.3 SD
Imputed scores +0.6 SD
Core sample (n = 300)
lnRMSSD
Posterior alpha
lnRMSSD and alpha index
Frontal alpha asymmetry
Recovery after the stressor
−0.02 0.00 0.02 0.04 0.06 0.08 0.10 0.12
Unstandardized indirect effect a × b (95% Monte Carlo CI)
b
Simple slope (95% CI)
0.1 0.0 −0.1 −0.2 −0.3 −0.4 −0.5
n.s. above 0.47 SD
−1 0 1 2
Challenge appraisal and mindset (SD)
c
Simple slope (95% CI)
0.1 0.0 −0.1 −0.2 −0.3 −0.4 −0.5
n.s. above 0.50 SD
−1 0 1 2
Emotion regulation (SD)